

CHENGHAO QIU

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EDUCATION

Texas A&M University

PhD @ Computer Science | GPA: 4.0 / 4.0

College Station, TX

Aug 2025 - Jun 2030 (Expected)

Tianjin University

Bachelor @ Computer Science | GPA: 3.75 / 4.0

Tianjin, China

Sep 2021 - Jun 2025

SKILLS AND INTERESTS

Program Language	Python SQL C++
ML Framework	PyTorch Huggingface
Tools & Liberties	Git Transformers Scikit-learn Pandas Numpy

RESEARCH EXPERIENCE

ZHOU LAB, Texas A&M University

Graduate Research Assistant, with Prof. Yi Zhou

Aug 2025 - present

- Revealed a **correctness–utility gap** in LLM in-context learning (ICL) by introducing **task-preserving exemplar perturbations**, demonstrating that even correct demonstrations can degrade accuracy across classification, reasoning, and math tasks, with larger drops in smaller models and harder settings.
- Developed **analog-friendly neural network** for efficient edge computing, including designing hardware-compatible NN architecture, developing noise-aware robustness training algorithms, and building a simulator to evaluate analog NN performance under realistic hardware conditions.

MAGICS LAB, Northwestern University

Undergraduate Research Assistant, with Prof. Han Liu

Apr 2024 - July 2025

- Proposed **GERM**, a fast and low-cost genomic foundation model for resource-constrained settings, by replacing standard attention with an outlier-free mechanism, improving fine-tuning performance by **37.98%** and quantization robustness by **64.34%** over the baseline.
- Developed the **GenoArmory** framework, the first unified benchmark for systematically evaluating the adversarial robustness of Genomic Foundation Models (GFMs), and introduced the **GenoAdv** dataset, achieving a **34.71%** increase in Defense Success Rate when used in training.

SELECTED PAPER

- [1] Haozheng Luo*, **Chenghao Qiu***, Maojiang Su, Zhihan Zhou, Zoe Mehta, Guo Ye, Jerry Yao-Chieh Hu, Han Liu. "Fast and Low-Cost Genomic Foundation Models via Outlier Removal." Forty-Second International Conference on Machine Learning (**ICML 2025**). [link](#)
- [2] Haozheng Luo*, Jiahao Yu*, Wenxin Zhang*, Jialong Li, **Chenghao Qiu**, Yimin Wang, Eric Jiang, Jerry Yao-Chieh Hu, Yan Chen, Binghui Wang, Xinyu Xing, Han Liu. "Decoupled alignment for robust plug-and-play adaptation." Third Conference on Language Modeling (**COLM 2026**). [link](#)

MANUSCRIPT IN PREPARATION

- [1] **Chenghao Qiu**, Chunli Peng, Yufeng Yang, Kuan-Hao Huang, Yi Zhou. "When Correct Demonstrations Hurt: Rethinking the Role of Exemplars in In-Context Learning." Manuscript in preparation for submission to NIPS 2026. [link](#)

*denotes equal contribution

- [2] Haozheng Luo*, **Chenghao Qiu***, Yimin Wang, Shang Wu, Jiahao Yu, Han Liu, Binghui Wang, Yan Chen. "GenoArmory: A Unified Evaluation Framework for Adversarial Attacks on Genomic Foundation Models." Manuscript in preparation for submission to NIPS 2026 ED Track. [link](#)
- [3] Haozheng Luo*, Haoran Dai*, Shaoyang Zhang, Xi Chen, Eric Hanchen Jiang, Yijiang Li, Jingyuan Huang, **Chenghao Qiu**, Chenwei Xu, Zhenyu Pan, Haotian Zhang, Binghui Wang, Yan Chen. "Attention Sinks and Outliers in Attention Residuals." Manuscript in preparation for submission to NIPS 2026. [link](#)